CHAPTER 1: PERIODIC TABLE - PERIODIC PROPERTIES AND VARIATIONS

Q1. An element in period 3, whose electron affinity is zero: [2023, 1 mark]

(a) Neon
(b) Sulphur
(c) Sodium
(d) Argon

Answer: (d) Argon

Explanation: Argon is a noble gas of period 3 whose electron affinity is zero due to stable electronic configuration.

Q2. An element with the largest atomic radius among the following is: [2023, 1 mark]

(a) Carbon
(b) Nitrogen
(c) Lithium
(d) Beryllium

Answer: (c) Lithium

Explanation: Atomic radius decreases across a period from left to right. Lithium is the leftmost element among these.

Q3. In the Periodic Table, elements of Period 3 are arranged in the increasing order of ionization potential as: [2022 Sem 1, 1 mark]

(a) B, N, Cl, Ar
(b) Mg, Si, S, Ar
(c) Ar, Si, S, Mg
(d) Si, Ar, Cl, Mg

Answer: (b) Mg, Si, S, Ar

Explanation: Ionization energy generally increases across period 3 because of increasing nuclear charge while shielding of outer electrons remains relatively same.

Q4. The most electronegative element from the following elements is: [2019, 1 mark]

(a) Magnesium
(b) Chlorine
(c) Aluminium
(d) Sulphur

Answer: (b) Chlorine

Q5. The element with highest ionisation potential is: [2020, 1 mark]

(a) Hydrogen
(b) Caesium
(c) Radon
(d) Helium

Answer: (d) Helium

Explanation: Helium has the highest ionization potential due to its small size and stable electronic configuration.

Q6. The non-metallic properties of elements from left to right in a Periodic Table: [2022 Sem 1, 1 mark]

(a) increases
(b) decreases
(c) remains same
(d) first increases and then decreases

Answer: (a) increases

Q7. If an element has low ionization potential, then it is likely to be a: [2022 Sem 1, 1 mark]

(a) metal
(b) metalloid
(c) non metal
(d) inert gas

Answer: (a) metal

Q8. The tendency of an atom to attract shared pair of electrons to itself when forming a chemical bond is known as: [2022 Sem 1, 1 mark]

(a) Electron affinity
(b) Electronegativity
(c) Ionization potential
(d) Nuclear charge

Answer: (b) Electronegativity

Q9. Alkaline earth metals have the same: [2022 Sem 1, 1 mark]

(a) number of valence electrons
(b) number of shells
(c) metallic property
(d) ionisation potential

Answer: (a) number of valence electrons

Q10. Among the period 2 elements, the element which has high electron affinity is: [2013, 1 mark]

(a) Lithium
(b) Carbon
(c) Fluorine
(d) Neon

Answer: (c) Fluorine

Q11. A group of elements in the Periodic Table are given below (Boron is the first member of the group and Thallium is the last): Boron, Aluminium, Gallium, Indium, Thallium. Which element has the most metallic character? [Undated, 5 marks]

(a) Boron
(b) Aluminium
(c) Indium
(d) Thallium

Answer: (d) Thallium

Q12. For the above group, which element would be expected to have the highest electronegativity? [Undated, 5 marks]

(a) Boron
(b) Aluminium
(c) Gallium
(d) Thallium

Answer: (a) Boron

CHAPTER 2: CHEMICAL BONDING

Q1. The compound that does not have a lone pair of electrons is: [2017, 1 mark]

(a) water
(b) ammonia
(c) carbon tetrachloride
(d) ammonium ion

Answer: (c) carbon tetrachloride

Q2. Which of the following has two sets of lone pair of electrons in them? [2022 Sem 1, 1 mark]

(a) Ammonia
(b) Methane
(c) Water
(d) Ammonium ion

Answer: (c) Water

Explanation: In water, oxygen has six valence electrons. It shares two electrons with two hydrogen atoms, leaving two lone pairs of electrons.

Q3. Bonding in this molecule can be understood to involve coordinate bonding: [2015, 1 mark]

(a) Carbon tetrachloride
(b) Hydrogen
(c) Hydrogen chloride
(d) Ammonium chloride

Answer: (d) Ammonium chloride

Q4. Among the following compounds, identify the compound which has all three bonds (ionic, covalent and coordinate bond): [2013, 1 mark]

(a) Ammonia
(b) Ammonium chloride
(c) Sodium hydroxide
(d) Calcium chloride

Answer: (b) Ammonium chloride

Q5. Type of bonding present in hydrogen chloride: [2022 Sem 1, 1 mark]

(a) metallic
(b) ionic
(c) covalent
(d) coordinate

Answer: (c) covalent

Q6. Which electron arrangement for the outer shell electrons in a covalent compound is correct? [2022 Sem 1, 1 mark]

(a) H:O:H
(b) H:N:H with lone pair
(c) H:N:H with lone pair (correct structure)
(d) H:C:H

Answer: (c) H:N:H with lone pair (correct structure for ammonia)

Q7. A compound with low boiling point is: [2020, 1 mark]

(a) Sodium chloride
(b) Calcium chloride
(c) Potassium chloride
(d) Carbon tetrachloride

Answer: (d) Carbon tetrachloride

Explanation: Carbon tetrachloride is a covalent compound with weak intermolecular forces, hence low boiling point.

Q8. The polar covalent compound in gaseous state that does not conduct electricity is: [2020, 1 mark]

(a) carbon tetrachloride
(b) ammonia
(c) methane
(d) hydrogen chloride

Answer: (b) ammonia

Q9. Which of the following is a common characteristic of a covalent compound? [2017, 1 mark]

(a) high melting point
(b) consists of molecules
(c) always soluble in water
(d) conducts electricity when in molten state

Answer: (b) consists of molecules

Q10. In covalent compounds, the bond is formed due to the _____ of electrons. [2011, 1 mark]

(a) transfer
(b) sharing
(c) donation
(d) acceptance

Answer: (b) sharing

CHAPTER 3: STUDY OF ACIDS, BASES AND SALTS

Q1. A salt formed by incomplete neutralisation of an acid by a base: [2022 Sem 1, 1 mark]

(a) Basic salt
(b) Acid salt
(c) Normal salt
(d) Complex salt

Answer: (b) Acid salt

Explanation: A salt formed by incomplete neutralization of an acid by a base is an Acid Salt. Example: $NaHSO_4$, $NaHCO_3$.

Q2. Which of the following is a tribasic acid? [2022 Sem 1, 1 mark]

(a) H_2SO_4
(b) $Al(OH)_3$
(c) H_3PO_4
(d) $Ca(OH)_2$

Answer: (c) H_3PO_4

Explanation: H_3PO_4 is a tribasic acid because it can donate all three protons attached to it.

Q3. The acidity of aluminium hydroxide is: [2022 Sem 1, 1 mark]

(a) 3
(b) 1
(c) 4
(d) 2

Answer: (a) 3

Explanation: $Al(OH)_3$ when reacts with any acid requires three acidic H^+ ions to neutralize, therefore its acidity is 3.

Q4. The acid which does not form acid salt by a basic radical: [2023, 1 mark]

(a) H_2CO_3
(b) H_3PO_4
(c) H_2SO_4
(d) CH_3COOH

Answer: (a) H_2CO_3

Explanation: It is a weak acid so does not participate in formation of acidic salts.

Q5. The products formed when an acid reacts with a base is: [2022 Sem 1, 1 mark]

(a) salt and hydrogen
(b) salt and oxygen
(c) salt and water
(d) salt and carbon dioxide

Answer: (c) salt and water

Q6. If the pH of a solution is 2, then the solution is a: [2022 Sem 1, 1 mark]

(a) strong acid
(b) strong alkali
(c) weak acid
(d) weak alkali

Answer: (a) strong acid

Q7. Higher the pH value of a solution, the more _____ it is. [2016, 1 mark]

(a) acidic
(b) alkaline
(c) neutral
(d) basic

Answer: (b) alkaline

Q8. A salt prepared by displacement reaction is: [2020, 1 mark]

(a) ferric chloride
(b) ferrous chloride
(c) silver chloride
(d) sodium chloride

Answer: (b) ferrous chloride

Q9. An alkali which completely dissociates into ions is: [2020, 1 mark]

(a) ammonium hydroxide
(b) calcium hydroxide
(c) lithium hydroxide
(d) sodium hydroxide

Answer: (d) sodium hydroxide

Q10. The indicator which changes to pink colour in an alkaline solution is: [2023, 1 mark]

(a) Blue Litmus
(b) Methyl Orange
(c) Red Litmus
(d) Phenolphthalein

Answer: (d) Phenolphthalein

CHAPTER 4: ANALYTICAL CHEMISTRY

Q1. The salt solution which does not react with ammonium hydroxide is: [2018, 1 mark]

(a) Calcium Nitrate
(b) Zinc Nitrate
(c) Lead Nitrate
(d) Copper Nitrate

Answer: (a) Calcium Nitrate

Q2. The metallic hydroxide which forms a deep inky blue solution with excess ammonium hydroxide solution is: [2023, 1 mark]

(a) $Fe(OH)_2$
(b) $Cu(OH)_2$

(c) $Ca(OH)_2$
(d) $Fe(OH)_3$

Answer: (b) $Cu(OH)_2$

Explanation: Cu^{2+} ions form deep inky blue solution with excess NH_4OH due to formation of $[Cu(NH_3)_4]^{2+}$.

Q3. Aqueous solution of cupric chloride forms a deep blue solution on addition of: [2022 Sem 1, 1 mark]

(a) dropwise sodium hydroxide
(b) excess sodium hydroxide
(c) dropwise ammonium hydroxide
(d) excess ammonium hydroxide

Answer: (d) excess ammonium hydroxide

Q4. A chloride which forms a precipitate that is soluble in excess of ammonium hydroxide is: [2017, 1 mark]

(a) Calcium chloride
(b) Ferrous chloride
(c) Ferric chloride
(d) Copper chloride

Answer: (d) Copper chloride

Q5. Silver chloride is a white precipitate that is soluble in excess of Ammonium hydroxide solution. [2023, 1 mark]

(a) True
(b) False
(c) Partially true
(d) None

Answer: (a) True

Q6. The colour of the precipitate formed after the addition of a small amount of sodium hydroxide solution to an aqueous solution of ferric chloride is: [2022 Sem 1, 1 mark]

(a) gelatinous white
(b) pale blue
(c) reddish brown
(d) dirty green

Answer: (c) reddish brown

Q7. Ammonium hydroxide solution when added to Solution B gives a white precipitate which does not dissolve in excess of ammonium hydroxide solution. The cation present is: [2023, 2 marks]

(a) Zn^{2+}
(b) Pb^{2+}
(c) Cu^{2+}
(d) Ca^{2+}

Answer: (b) Pb^{2+}

Q8. Sodium hydroxide solution when added to Solution C gives a white precipitate which is insoluble in excess of sodium hydroxide solution. The cation present is: [2023, 2 marks]

(a) Zn^{2+}
(b) Pb^{2+}
(c) Ca^{2+}
(d) Al^{3+}

Answer: (c) Ca^{2+}

CHAPTER 5: MOLE CONCEPT AND STOICHIOMETRY

Q1. If Relative Molecular Mass of Butane (C_4H_{10}) is 58 then, its vapour density will be: [2022 Sem 1, 1 mark]

(a) 58
(b) 29
(c) 32
(d) 16

Answer: (b) 29

Explanation: Molecular mass = 2 × vapour density; 58 = 2 × VD; VD = 29.

Q2. The vapour density of CH_3OH is (At. Wt. C = 12, H = 1, O = 16): [2023, 1 mark]

(a) 32
(b) 18
(c) 16
(d) 34

Answer: (c) 16

Explanation: Molecular mass = 12 + 3×1 + 16 + 1 = 32; VD = 32/2 = 16.

Q3. The percentage of hydrogen present in NaOH is (Relative Molecular Mass of NaOH = 40): [2022 Sem 1, 1 mark]

(a) 2.5
(b) 25
(c) 0.25
(d) 0.025

Answer: (a) 2.5

Explanation: % composition of Hydrogen = (1/40) × 100 = 2.5%.

Q4. If the empirical mass of the formula PQ_2 is 10 and the Relative Molecular Mass is 30, then the molecular formula will be: [2022 Sem 1, 1 mark]

(a) PQ_2
(b) P_3Q_2
(c) P_6Q_3
(d) P_3Q_6

Answer: (d) P_3Q_6

Explanation: n = Molecular mass / Empirical mass = 30/10 = 3; Molecular formula = $(PQ_2)_3 = P_3Q_6$.

Q5. If the empirical formula of a compound is CH and its vapour density is 13, then its molecular formula will be (At. Wt. C = 12, H = 1): [2022 Sem 1, 1 mark]

(a) CH
(b) C_2H_2
(c) C_4H_4
(d) C_3H_3

Answer: (b) C_2H_2

Explanation: Molecular weight = 2 × VD = 26; Empirical formula mass = 13; n = 26/13 = 2; Molecular formula = $(CH)_2 = C_2H_2$.

Q6. The number of moles in 11 gm of nitrogen gas is (atomic mass of N = 14): [2020, 1 mark]

(a) 0.39
(b) 0.49
(c) 0.29
(d) 0.59

Answer: (a) 0.39

Explanation: Molar mass of N_2 = 28; Number of moles = 11/28 = 0.39.

Q7. The mass of 11.2 litres of a certain gas at s.t.p. is 24 g. The gram molecular mass of the gas is: [2017, 1 mark]

(a) 24
(b) 48
(c) 12
(d) 36

Answer: (b) 48

Explanation: 11.2 L at STP = 0.5 mole; 0.5 mole = 24 g; 1 mole = 48 g.

Q8. A gas cylinder can hold 1 kg of hydrogen at room temperature and pressure. The number of moles of hydrogen present is: [2017, 1 mark]

(a) 500
(b) 250
(c) 1000
(d) 125

Answer: (a) 500

Explanation: Molar mass of H_2 = 2 g/mol; Number of moles = 1000/2 = 500.

Q9. The volume occupied by 80 g of carbon dioxide at STP is: [2020, 1 mark]

(a) 22.4 L
(b) 44.8 L
(c) 40.72 L
(d) 11.2 L

Answer: (c) 40.72 L

Explanation: Molar mass CO_2 = 44 g; 44 g occupies 22.4 L; 80 g occupies (22.4 × 80)/44 = 40.72 L.

Q10. The number of molecules in 4.4 gm of CO_2 is: [2020, 1 mark]

(a) 6.022×10^{23}
(b) 6.022×10^{22}
(c) 3.011×10^{23}
(d) 3.011×10^{22}

Answer: (b) 6.022×10^{22}

Explanation: 44 g CO_2 contains 6.022×10^{23} molecules; 4.4 g contains $(4.4/44) \times 6.022 \times 10^{23} = 6.022 \times 10^{22}$.

CHAPTER 6: ELECTROLYSIS

Q1. Identify one statement that holds true for electrolysis of molten lead bromide: [2022 Sem 1, 1 mark]

(a) Silver grey metal deposits at the anode.
(b) Temperature is not maintained during the electrolysis.
(c) Brown vapours of bromine are obtained at the anode.
(d) Electrolyte contains H^+ ions along with Pb^{2+} ions.

Answer: (c) Brown vapours of bromine are obtained at the anode.

Explanation: At anode: $2Br^- \rightarrow Br_2 + 2e^-$ (Reddish brown vapours).

Q2. The inert electrode used in the electrolysis of acidified water is: [2020, 1 mark]

(a) Nickel
(b) Platinum
(c) Copper
(d) Silver

Answer: (b) Platinum

Q3. During the electrolysis of molten lead bromide, which of the following takes place? [2008, 1 mark]

(a) Lead is deposited at the anode
(b) Bromine is deposited at the cathode
(c) Lead is deposited at the cathode
(d) Bromine is deposited at the cathode

Answer: (c) Lead is deposited at the cathode

Q4. Which of the following ions will readily discharge at the anode during the electrolysis of acidified water? [2022 Sem 1, 1 mark]

(a) OH^-
(b) SO_4^{2-}
(c) Cl^-
(d) H^+

Answer: (a) OH^-

Q5. The ion which could be discharged most readily during electrolysis is: [2023, 2 marks]

(a) Fe^{2+}
(b) Cu^{2+}
(c) Na^+
(d) Mg^{2+}

Answer: (b) Cu^{2+}

Q6. If a solution of an electrolyte mixture has calcium ions, cupric ions, zinc ions and magnesium ions, which of these ions would you see preferentially discharged at the cathode? [2022 Sem 1, 1 mark]

(a) Calcium ions
(b) Zinc ions

(c) Cupric ions
(d) Magnesium ions

Answer: (c) Cupric ions

Q7. Solid sodium chloride does not conduct electricity as: [2022 Sem 1, 1 mark]

(a) The strength of the bond is weak.
(b) It contains free ions.
(c) It does not contain any free ions.
(d) It contains free ions as well as molecules.

Answer: (c) It does not contain any free ions.

Q8. Which of the following reactions takes place at the anode during the electroplating of an article with silver? [2023, 1 mark]

(a) $Ag - 1e^- \rightarrow Ag^+$
(b) $Ag + 1e^- \rightarrow Ag^-$
(c) $Ag - 1e^- \rightarrow Ag$
(d) None of the above

Answer: (a) $Ag - 1e^- \rightarrow Ag^+$

Q9. The electrolysis of acidified water is an example of: [2018, 1 mark]

(a) Reduction
(b) Oxidation
(c) Redox reaction
(d) Synthesis

Answer: (c) Redox reaction

Q10. Which statement about conduction of electricity is correct? [2022 Sem 1, 1 mark]

(a) Electricity is conducted in aqueous solution by electrons.
(b) Electricity is conducted in a metal wire by ions.
(c) Electricity is conducted in a molten electrolyte by electrons.
(d) Electricity is conducted in an acid solution by ions.

Answer: (d) Electricity is conducted in an acid solution by ions.

CHAPTER 7: METALLURGY

Q1. The ore of Aluminium is: [2022 Sem 2, 1 mark]

(a) Calamine
(b) Haematite
(c) Magnetite
(d) Cryolite

Answer: (d) Cryolite

Q2. The compound that is not an ore of aluminium: [2023, 1 mark]

(a) Cryolite
(b) Corundum
(c) Fluorspar
(d) Bauxite

Answer: (c) Fluorspar

Explanation: Fluorspar is CaF_2, not an ore of aluminium.

Q3. The main ore used for the extraction of iron is: [2014, 1 mark]

(a) Haematite
(b) Calamine

(c) Bauxite
(d) Cryolite

Answer: (a) Haematite

Q4. Heating an ore in a limited supply of air or in the absence of air at a temperature just below its melting point is known as: [2014, 1 mark]

(a) Smelting
(b) Ore dressing
(c) Calcination
(d) Bessemerisation

Answer: (c) Calcination

Q5. The process by which sulphide ore is concentrated is: [2018, 1 mark]

(a) Roasting
(b) Calcination
(c) Froth flotation
(d) Magnetic separation

Answer: (c) Froth flotation

Q6. An ore of zinc containing its sulphide is: [2017, 1 mark]

(a) Zinc blende
(b) Calamine
(c) Zincite
(d) Galena

Answer: (a) Zinc blende

Q7. The ore that can be concentrated by using magnetic separation: [2025, 1 mark]

(a) Corundum
(b) Haematite
(c) Calamine
(d) Bauxite

Answer: (b) Haematite

Q8. Substance which helps to lower the fusion point of the mixture in Hall Heroult Process: [2022 Sem 2, 1 mark]

(a) Coke
(b) Concentrated sodium hydroxide
(c) Fluorspar
(d) Concentrated potassium hydroxide

Answer: (c) Fluorspar

Q9. In the extraction of aluminium, the electrolyte consists of: [2011, 1 mark]

(a) Alumina only
(b) Cryolite only
(c) Alumina + Cryolite + Fluorspar
(d) Bauxite only

Answer: (c) Alumina + Cryolite + Fluorspar

Q10. The chemical formula of cryolite is: [2019, 1 mark]

(a) $NaAlO_2$
(b) Na_3AlF_6
(c) $Al_2O_3 \cdot 2H_2O$
(d) CaF_2

Answer: (b) Na_3AlF_6

CHAPTER 8: STUDY OF COMPOUNDS - HYDROGEN CHLORIDE

Q1. Hydrogen chloride gas is not collected over water, as: [2022 Sem 2, 1 mark]

(a) It is highly soluble in water.
(b) It is less soluble in water.
(c) It is lighter than air.
(d) It is heavier than air.

Answer: (a) It is highly soluble in water.

Q2. The drying agent used to dry HCl gas is: [2019, 1 mark]

(a) Conc. H_2SO_4
(b) ZnO
(c) Al_2O_3
(d) CaO

Answer: (a) Conc. H_2SO_4

Q3. In the laboratory preparation, HCl gas is dried by passing through: [2023, 1 mark]

(a) dilute nitric acid
(b) concentrated sulphuric acid
(c) dilute sulphuric acid
(d) acidified water

Answer: (b) concentrated sulphuric acid

Q4. Hydrogen chloride gas cannot be dried over quick lime because: [2023, 2 marks]

(a) Quick lime absorbs HCl
(b) Quick lime reacts with HCl

(c) Quick lime is basic and HCl is acidic
(d) Both (b) and (c)

Answer: (d) Both (b) and (c)

Explanation: HCl is acidic and quicklime (CaO) is basic, so they react: $2HCl + CaO \rightarrow CaCl_2 + H_2O$.

Q5. When sodium chloride is heated with concentrated sulphuric acid below 200°C, one of the products formed is: [2019, 1 mark]

(a) sodium hydrogen sulphate
(b) sodium sulphate
(c) chlorine
(d) sodium chloride

Answer: (a) sodium hydrogen sulphate

Q6. The acid which is least volatile is: [2022 Sem 2, 1 mark]

(a) Hydrochloric acid
(b) Nitric acid
(c) Dilute sulphuric acid
(d) Concentrated sulphuric acid

Answer: (d) Concentrated sulphuric acid

Q7. The property exhibited by concentrated sulphuric acid when it is used to prepare hydrogen chloride gas from potassium chloride: [2023, 1 mark]

(a) Dehydrating property
(b) Drying property
(c) Oxidizing property
(d) Non-volatile acid property

Answer: (d) Non-volatile acid property

Q8. The salt formed when concentrated sulphuric acid reacts with KNO_3 above 200°C: [2023, 1 mark]

(a) K_2SO_4
(b) K_2SO_3
(c) $KHSO_4$
(d) $KHSO_3$

Answer: (a) K_2SO_4

CHAPTER 9: STUDY OF COMPOUNDS - AMMONIA

Q1. An aqueous solution of ammonia is: [2022 Sem 2, 1 mark]

(a) Neutral
(b) Acidic
(c) Basic
(d) Amphoteric

Answer: (c) Basic

Q2. Ammonia reacts with excess chlorine to form: [2019, 1 mark]

(a) nitrogen
(b) nitrogen trichloride
(c) ammonium chloride
(d) nitric acid

Answer: (b) nitrogen trichloride

Q3. Ammonia gas is not collected over water because: [2023, 2 marks]

(a) It is lighter than air
(b) It is highly soluble in water
(c) It reacts with water
(d) It is heavier than air

Answer: (b) It is highly soluble in water

Q4. The catalyst used to oxidize ammonia into nitric oxide is: [2020, 1 mark]

(a) Iron
(b) Platinum
(c) Vanadium pentoxide
(d) Nickel

Answer: (b) Platinum

Q5. Ammonia can be obtained by adding water to: [2008, 1 mark]

(a) Ammonium chloride
(b) Ammonium nitrate
(c) Magnesium nitride
(d) Magnesium nitrate

Answer: (c) Magnesium nitride

Q6. An alkaline gas which produces dense white fumes when reacted with hydrogen chloride gas is: [2008, 1 mark]

(a) Ammonia
(b) Oxygen
(c) Nitrogen
(d) Hydrogen

Answer: (a) Ammonia

Q7. The catalyst used in the Haber process for the manufacture of ammonia is: [2012, 1 mark]

(a) Platinum
(b) Iron
(c) Vanadium pentoxide
(d) Nickel

Answer: (b) Iron

Q8. Ammonia burns in oxygen with a: [2015, 1 mark]

(a) blue flame
(b) yellow flame
(c) greenish yellow flame
(d) colorless flame

Answer: (c) greenish yellow flame

Q9. The compound formed when excess ammonia gas reacts with chlorine is: [2022 Sem 2, 2 marks]

(a) NH_4Cl
(b) NCl_3
(c) N_2
(d) HCl

Answer: (b) NCl_3

Q10. In the laboratory preparation of ammonia, the mixture used is: [2017, 1 mark]

(a) $NH_4Cl + Ca(OH)_2$
(b) $NH_4Cl + NaOH$
(c) $(NH_4)_2SO_4 + CaO$
(d) $NH_4NO_3 + Ca(OH)_2$

Answer: (a) $NH_4Cl + Ca(OH)_2$

CHAPTER 10: STUDY OF COMPOUNDS - NITRIC ACID

Q1. The acid which can produce carbon from cane sugar is: [2020, 1 mark]

(a) Concentrated Hydrochloric acid
(b) Concentrated Nitric acid
(c) Concentrated Sulphuric acid
(d) Concentrated Acetic acid

Answer: (c) Concentrated Sulphuric acid

Q2. A dilute mineral acid which forms a white precipitate when treated with barium chloride solution is: [2013, 1 mark]

(a) HCl
(b) HNO_3
(c) H_2SO_4
(d) CH_3COOH

Answer: (c) H_2SO_4

Q3. The acid which cannot act as an oxidizing agent is: [2020, 1 mark]

(a) conc. H_2SO_4
(b) conc. HNO_3
(c) conc. HCl
(d) dilute H_2SO_4

Answer: (c) conc. HCl

Q4. The acid formed when sulphur reacts with concentrated nitric acid is: [2022 Sem 2, 3 marks]

(a) Sulphurous acid
(b) Sulphuric acid
(c) Nitrous acid
(d) Hydrosulphuric acid

Answer: (b) Sulphuric acid

Q5. The acid formed when potassium nitrate reacts with a least volatile acid is: [2022 Sem 2, 3 marks]

(a) HCl
(b) H_2SO_4
(c) HNO_3
(d) H_3PO_4

Answer: (c) HNO_3

Q6. The nitrate which on thermal decomposition leaves behind a residue which is yellow when hot and white when cold: [2023, 1 mark]

(a) Lead nitrate
(b) Ammonium nitrate
(c) Copper nitrate
(d) Zinc nitrate

Answer: (a) Lead nitrate

Q7. When copper reacts with concentrated nitric acid, the gas evolved is: [2011, 1 mark]

(a) NO
(b) NO_2
(c) N_2O
(d) N_2

Answer: (b) NO_2

Q8. The brown ring test is used for the detection of: [2011, 1 mark]

(a) CO_3^{2-}
(b) NO_3^-
(c) SO_3^{2-}
(d) Cl^-

Answer: (b) NO_3^-

Q9. Concentrated nitric acid appears yellow when it is left for a while in a glass bottle because: [2020, 1 mark]

(a) It absorbs moisture
(b) It decomposes to give NO_2 which dissolves in it
(c) It reacts with glass
(d) It forms HNO_2

Answer: (b) It decomposes to give NO_2 which dissolves in it

Q10. In the laboratory preparation of nitric acid, the apparatus used is made entirely of glass because: [2015, 1 mark]

(a) Nitric acid is corrosive
(b) Nitric acid reacts with rubber and cork
(c) Glass is cheap
(d) Glass is transparent

Answer: (b) Nitric acid reacts with rubber and cork

CHAPTER 11: STUDY OF COMPOUNDS - SULPHURIC ACID

Q1. Vanadium pentoxide is used as a catalyst in the preparation of: [2022 Sem 2, 1 mark]

(a) Nitrogen gas
(b) Nitrogen dioxide gas
(c) Sulphur trioxide gas
(d) Carbon dioxide gas

Answer: (c) Sulphur trioxide gas

Q2. The catalyst used in the Contact Process is: [2018, 1 mark]

(a) Copper
(b) Iron
(c) Vanadium pentaoxide
(d) Manganese dioxide

Answer: (c) Vanadium pentaoxide

Q3. The property of concentrated sulphuric acid when it is used to prepare hydrogen chloride gas from potassium chloride is: [2023, 1 mark]

(a) Dehydrating property
(b) Drying property
(c) Oxidizing property
(d) Non-volatile acid property

Answer: (d) Non-volatile acid property

Q4. The gas formed when calcium bisulphite reacts with dilute HNO_3: [2022 Sem 2, 1 mark]

(a) Sulphur trioxide
(b) Hydrogen
(c) Sulphur dioxide
(d) Hydrogen sulphide

Answer: (c) Sulphur dioxide

Q5. In the reaction $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$, sulphuric acid acts as: [2025, 1 mark]

(a) Non-volatile acid
(b) Dibasic acid
(c) Oxidising agent
(d) Reducing agent

Answer: (c) Oxidising agent

Q6. The acid which is least volatile is: [2022 Sem 2, 1 mark]

(a) Hydrochloric acid
(b) Nitric acid
(c) Dilute sulphuric acid
(d) Concentrated sulphuric acid

Answer: (d) Concentrated sulphuric acid

Q7. A substance added to water to manufacture sulphuric acid in Contact process is: [2022 Sem 2, 2 marks]

(a) SO_2
(b) SO_3
(c) Oleum
(d) $H_2S_2O_7$

Answer: (c) Oleum

Q8. Concentrated sulphuric acid acts as a dehydrating agent when it reacts with: [2015, 1 mark]

(a) Copper
(b) Sugar
(c) Sulphur
(d) Sodium chloride

Answer: (b) Sugar

Q9. The salt formed when concentrated sulphuric acid reacts with KNO_3 below 200°C: [2023, 1 mark]

(a) K_2SO_4
(b) K_2SO_3
(c) $KHSO_4$
(d) $KHSO_3$

Answer: (c) $KHSO_4$

Q10. When dilute sulphuric acid reacts with iron sulphide, the gas evolved is: [2011, 1 mark]

(a) Hydrogen sulphide
(b) Sulphur dioxide
(c) Sulphur trioxide
(d) Vapour of sulphuric acid

Answer: (a) Hydrogen sulphide

CHAPTER 12: ORGANIC CHEMISTRY - HYDROCARBONS

Q1. Unsaturated hydrocarbons undergo: [2024, 1 mark]

(a) Addition reaction
(b) Substitution reaction
(c) Oxidation reaction
(d) Redox reaction

Answer: (a) Addition reaction

Q2. The organic compound having a triple carbon-carbon covalent bond is: [2020, 1 mark]

(a) C_3H_4
(b) C_3H_6
(c) C_3H_8
(d) C_4H_{10}

Answer: (a) C_3H_4

Q3. The organic compound which undergoes substitution reaction is: [2018, 1 mark]

(a) C_2H_2
(b) C_2H_4
(c) $C_{10}H_{18}$
(d) C_2H_6

Answer: (d) C_2H_6

Q4. A hydrocarbon which is a greenhouse gas is: [2019, 1 mark]

(a) Acetylene
(b) Ethylene
(c) Ethane
(d) Methane

Answer: (d) Methane

Q5. The general formula of hydrocarbons with single covalent bonds is: [2023, 1 mark]

(a) $C\square H_2\square_{+2}$
(b) $C\square H_2\square$
(c) $C\square H_2\square_{-2}$
(d) $C\square H_2\square_{-6}$

Answer: (a) $C\square H_2\square_{+2}$

Q6. An example of a cyclic organic compound is: [2023, 1 mark]

(a) Propene
(b) Pentene
(c) Butene
(d) Benzene

Answer: (d) Benzene

Q7. The unsaturated hydrocarbons having general formula C_nH_{2n-2} are: [2022 Sem 2, 2 marks]

(a) Alkanes
(b) Alkenes
(c) Alkynes
(d) Aromatics

Answer: (c) Alkynes

Q8. The IUPAC name of acetylene is: [2014, 1 mark]

(a) Propane
(b) Propyne
(c) Ethene
(d) Ethyne

Answer: (d) Ethyne

Q9. The formation of 1,2-dibromoethane from ethene and bromine is an example of: [2008, 1 mark]

(a) Substitution
(b) Dehydration
(c) Dehydrohalogenation
(d) Addition

Answer: (d) Addition

Q10. The hydrocarbon formed when sodium propanoate and soda lime are heated together: [2023, 1 mark]

(a) Methane
(b) Ethane
(c) Ethene
(d) Propane

Answer: (b) Ethane

CHAPTER 13: ORGANIC CHEMISTRY - ALCOHOLS AND CARBOXYLIC ACIDS

Q1. The functional group present in acetic acid is: [2011, 1 mark]

(a) Ketonic C=O
(b) Hydroxyl -OH
(c) Aldehyde -CHO
(d) Carboxylic -COOH

Answer: (d) Carboxylic -COOH

Q2. The IUPAC name of formic acid: [2022 Sem 2, 1 mark]

(a) Propanoic acid
(b) Methanoic acid
(c) Ethanoic acid
(d) Butanoic acid

Answer: (b) Methanoic acid

Q3. Conversion of ethanol to ethene by the action of concentrated sulphuric acid involves: [2024, 1 mark]

(a) Dehydration
(b) Dehydrogenation
(c) Dehydrohalogenation
(d) Hydrolysis

Answer: (a) Dehydration

Q4. Reaction of an alcohol with a carboxylic acid in the presence of concentrated H_2SO_4 is termed as: [2024, 1 mark]

(a) Halogenation
(b) Esterification
(c) Hydrogenation
(d) Dehydrohalogenation

Answer: (b) Esterification

Q5. The organic compound obtained as the end product of the fermentation of sugar solution is: [2009, 1 mark]

(a) Methane
(b) Ethanol
(c) Acetic acid
(d) Ethene

Answer: (b) Ethanol

Q6. The IUPAC name of dimethyl ether is: [2018, 1 mark]

(a) Ethoxy methane
(b) Methoxy methane
(c) Methoxy ethane
(d) Ethoxy ethane

Answer: (b) Methoxy methane

Q7. The compound that reacts with acetic acid to form ethyl ethanoate is: [2019, 4 marks]

(a) Methanol
(b) Ethanol
(c) Propanol
(d) Butanol

Answer: (b) Ethanol

Q8. The organic compound which when solidified forms an ice-like mass is: [2019, 3 marks]

(a) Ethanol
(b) Methanol
(c) Acetic acid
(d) Acetone

Answer: (c) Acetic acid

Q9. The compound formed by complete chlorination of ethyne is: [2019, 4 marks]

(a) Acetylene tetrachloride
(b) Ethyl chloride
(c) Vinyl chloride
(d) Ethylene dichloride

Answer: (a) Acetylene tetrachloride

Q10. Ethanoic acid reacts with ethanol in the presence of concentrated H_2SO_4 to form: [2012, 1 mark]

(a) Ethene
(b) Ethyl ethanoate
(c) Ethane
(d) Ethoxy ethane

Answer: (b) Ethyl ethanoate

CHAPTER 14: ALLOYS

Q1. Copper, Zinc, and Tin are the metals alloyed to form: [2024, 1 mark]

(a) Duralumin
(b) Brass
(c) Bronze
(d) Solder

Answer: (d) Solder

Q2. Brass is an alloy of: [2008, 1 mark]

(a) Copper and tin
(b) Copper and zinc
(c) Zinc and lead
(d) Lead and tin

Answer: (b) Copper and zinc

Q3. The two main metals in Bronze are: [2016, 1 mark]

(a) Copper and zinc
(b) Copper and lead
(c) Copper and nickel
(d) Copper and tin

Answer: (d) Copper and tin

Q4. An alloy of lead and tin that is used in electrical circuits is: [2017, 1 mark]

(a) Solder
(b) Brass
(c) Bronze
(d) Duralumin

Answer: (a) Solder

Q5. The reason for using Aluminium in the alloy duralumin is: [2019, 1 mark]

(a) Aluminium is brittle.
(b) Aluminium gives strength.
(c) Aluminium brings lightness.
(d) Aluminium lowers melting point.

Answer: (c) Aluminium brings lightness.

Q6. The alloy containing Cu and Zn is hard, silvery and is used in decorative articles: [2012, 1 mark]

(a) Duralumin
(b) Brass
(c) Bronze
(d) Solder

Answer: (b) Brass

Q7. This is not an alloy of copper: [2015, 1 mark]

(a) Brass
(b) Bronze

(c) Solder
(d) Duralumin

Answer: (d) Duralumin

Q8. An alloy used to make statues is: [2020, 1 mark]

(a) bronze
(b) brass
(c) fuse metal
(d) stainless steel

Answer: (a) bronze

Q9. The alloy of aluminium, magnesium, manganese and copper is: [2023, 2 marks]

(a) Brass
(b) Bronze
(c) Duralumin
(d) Solder

Answer: (c) Duralumin

Q10. The alloy of iron, nickel, chromium and carbon is: [2023, 2 marks]

(a) Stainless steel
(b) Duralumin
(c) Brass
(d) Bronze

Answer: (a) Stainless steel

CHAPTER 15: PRACTICAL CHEMISTRY

Q1. Which gas decolourises potassium permanganate $KMnO_4$ solution? [2025, 1 mark]

(a) Sulphur dioxide
(b) Ammonia
(c) Hydrogen chloride
(d) Carbon dioxide

Answer: (a) Sulphur dioxide

Q2. Rotten egg smell is due to the liberation of: [2024, 1 mark]

(a) HCl gas
(b) H_2S gas
(c) Cl_2 gas
(d) SO_2 gas

Answer: (b) H_2S gas

Q3. Ammonia gas is collected by downward displacement of air since ammonia is: [2024, 1 mark]

(a) very slightly soluble in water
(b) heavier than air
(c) lighter than air
(d) insoluble in water

Answer: (c) lighter than air

Q4. The salt solution which does not react with ammonium hydroxide is: [2018, 1 mark]

(a) Calcium Nitrate
(b) Zinc Nitrate
(c) Lead Nitrate
(d) Copper Nitrate

Answer: (a) Calcium Nitrate

Q5. To increase the pH value of a neutral solution, we should add: [2017, 1 mark]

(a) an acid
(b) an acid salt
(c) an alkali
(d) a salt

Answer: (c) an alkali

Q6. Anhydrous iron(III) chloride is prepared by: [2017, 1 mark]

(a) direct combination
(b) simple displacement
(c) decomposition
(d) neutralization

Answer: (a) direct combination

Q7. The aqueous solution that contains both ions and molecules: [2022 Sem 1, 1 mark]

(a) sulphuric acid
(b) nitric acid
(c) acetic acid
(d) hydrochloric acid

Answer: (c) acetic acid

Q8. The basic oxide which is an alkali: [2022 Sem 1, 1 mark]

(a) Copper oxide
(b) Sodium oxide
(c) Ferric oxide
(d) Zinc oxide

Answer: (b) Sodium oxide

Q9. Which of the following will act as a non-electrolyte? [2013, 1 mark]

(a) Liquid carbon tetrachloride
(b) Acetic acid
(c) Sodium hydroxide aqueous solution
(d) Potassium chloride aqueous solution

Answer: (a) Liquid carbon tetrachloride

Q10. Which of the following will not produce an acid when made to react with water? [2013, 1 mark]

(a) Carbon monoxide
(b) Carbon dioxide
(c) Nitrogen dioxide
(d) Sulphur trioxide

Answer: (a) Carbon monoxide

CHAPTER 16: MOLE CONCEPT - EMPIRICAL AND MOLECULAR FORMULA

Q1. An organic compound has a vapour density of 22. The molecular formula of the organic compound is (Atomic weight: C = 12, H = 1): [2025, 1 mark]

(a) CH_4
(b) C_2H_4
(c) C_2H_6
(d) C_3H_8

Answer: (d) C_3H_8

Explanation: Molecular weight = 2 × VD = 44; C_3H_8 = 36 + 8 = 44.

Q2. The vapour density of carbon dioxide (C = 12, O = 16): [2012, 1 mark]

(a) 32
(b) 16
(c) 44
(d) 22

Answer: (d) 22

Explanation: Molecular mass CO_2 = 44; VD = 44/2 = 22.

Q3. A compound has the percentage composition by mass: carbon 14.4%, hydrogen 1.2% and chlorine 84.5%. The empirical formula of this compound is: [2008, 6 marks]

(a) CHCl
(b) $CHCl_2$
(c) CH_2Cl
(d) $C_2H_2Cl_2$

Answer: (b) $CHCl_2$

Q4. The empirical formula of a compound is CH and it has a vapour density of 13. The molecular formula of the compound is: [2015, 1 mark]

(a) C_2H_2
(b) C_2H_4
(c) C_2H_6
(d) C_6H_6

Answer: (a) C_2H_2

Q5. The empirical formula of C_8H_{18} is: [2023, 3 marks]

(a) C_4H_9
(b) C_2H_4

(c) C_8H_{18}
(d) CH_2

Answer: (a) C_4H_9

Q6. The Empirical formula of an organic compound is $CHCl_2$. If its relative molecular mass is 168, its molecular formula is (At. Wt. C = 12, H = 1, Cl = 35.5): [2023, 2 marks]

(a) $CHCl_2$
(b) $C_2H_2Cl_4$
(c) $C_2H_2Cl_2$
(d) $C_3H_3Cl_6$

Answer: (b) $C_2H_2Cl_4$

Q7. A compound made up of two elements X and Y has an empirical formula X_2Y. If the atomic weight of X is 10 and that of Y is 5 and the compound has a vapour density 25, its molecular formula is: [2014, 5 marks]

(a) X_2Y
(b) X_4Y_2
(c) X_2Y_4
(d) X_6Y_3

Answer: (b) X_4Y_2

Q8. A compound X consists of 4.8% carbon and 95.2% bromine by mass. The empirical formula of this compound is (C = 12, Br = 80): [Undated, 5 marks]

(a) CBr
(b) CBr_2
(c) CBr_3
(d) C_2Br_3

Answer: (c) CBr_3

Q9. The percentage of nitrogen and oxygen in ammonium nitrate (Relative molecular mass = 80) are: [2010, 5 marks]

(a) N = 35%, O = 60%
(b) N = 35%, O = 65%
(c) N = 30%, O = 60%
(d) N = 40%, O = 50%

Answer: (a) N = 35%, O = 60%

Q10. The percentage of water of crystallization in $CuSO_4 \cdot 5H_2O$ is (H = 1, O = 16, S = 32, Cu = 64): [2017, 1 mark]

(a) 36%
(b) 32%
(c) 28%
(d) 24%

Answer: (a) 36%